卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-CET-DD-EC-0405 49-B60011S-D0405

第 1 页 PAGE 共 1 页 OF

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 04 卷 第 05 册
Electrical SPECIALITY VOLUME BOOK

卷册 名称 交流滤波场围栏内辅助地网图
Volume name Grounding Grid Inside AC Filter Bank Fence

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图 纸 DRAWING NOS. 6 张	打印件 PRINTED COPIES NOS. / 本

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	49-B60011S-D0405-001	卷册施工说明 Description of Construction	A	1		
2	49-B60011S-D0405-002	SC并联电容器小组地网布置图 SC Filter Bank Grounding Grid Plan	A	1		
3	49-B60011S-D0405-003	HP12/24交流滤波器小组地网布置图 HP12/24 AC Filter Bank Grounding Grid Plan	A	1		
4	49-B60011S-D0405-004	接地体敷设详图 Detail Drawing of Grounding Conductor Laying	A	1		
5	49-B60011S-D0405-005	500kV交流滤波器区域围栏平面布置图 Layout of 500kV AC Filter Bank Fence	A	1		
6	49-B60011S-D0405-006	材料汇总表 Tables of Material	A	1		

注： 本工程初步设计尚未审定，未通过NTDC审核，图纸存在修改的可能，本册图纸施工前请联系CET确认。

The preliminary design of this project has not been approved and has not been approved by NTDC.

The modification is possible in this diagram. Please contact CET for confirmation before construction.

卷册施工说明

一、工程概况及设计原则

1、项目名称：巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程/拉合尔±660kV换流站；

2、建设单位：国家电网/中国电力技术装备有限公司

3、设计规范，包含但不限于：IEEE 80-2013 IEEE Guide for Safety in AC Substation Grounding。

二、设计及施工接口

1、本册图纸适用于500kV交流滤波场围栏内辅助地网的布置，包含围栏内的垂直接地极布置。围栏内垂直接地极主要用于避雷器用集中接地装置和中性点母线的加强接地。

2、围栏内辅助地网引出至相邻围栏地网或至场区主地网。围栏外场区主地网的布置由华东院负责设计。

3、围栏内的设备（设施）、围栏、端子箱等的接地设计详见各围栏设备安装卷册。

三、施工注意事项

1、500kV滤波场围栏内辅助地网采用TJ-120铜绞线，敷设深度为0.8m，其连接处采用放热焊接。过道路和电缆沟的接地体应敷设在垫层以下的老土/回填土中。

2、围栏内垂直接地极采用Ø17.2mm，L=3000mm铜覆钢棒，任意两个垂直接地极距离宜大于6m，所有垂直接地极必须与辅助地网连接，连接处采用放热焊接。

3、横向和纵向接地体相互交叉处应焊接在一起，铜或者铜覆钢接地体采用放热焊接方式。辅助地网不应与设备基础、构支架基础重合，遇到基础由侧面绕行敷设。各围栏内辅助地网标示尺寸供参考，施工过程中可根据现场实际情况微调。

4、围栏内所有避雷器均应设集中接地装置，集中接地装置就近与辅助接地网连接。围栏内所有中性点母线需加强接地，母线接地引下线增设垂直接地极，就近与辅助接地网连接。

5、围栏内所有电气设备外壳和底座、金属支架、金属围栏、金属门、电缆护管等均应与辅助地网可靠焊接。上述围栏内设备接地用材料及示意图详见各围栏设备安装卷册。

四、其他

本卷册要求包含但不限于上述内容，未尽事宜，请遵照国际标准、国家标准、行业标准、施工验收规范等文件执行。

1. Project Overview and Standard

a. Project name: ±660kV HVDC Project from Matiari to Lahore in Pakistan/±660kV Lahore Converter Station.

b. Development organization: State Grid/China Electric Power Equipment and Technology CO. LTD.

c. Standard(including but not limited): IEEE 80-2013 IEEE Guide for Safety in AC Substation Grounding.

2. Design and Construction Scope

a. Drawings in this volume are applicable to the layout of grounding grid inside 500kV AC filter bank fences, including vertical grounding poles in the fences. The vertical earthing pole in the fence is mainly used for the reinforced grounding of the centralized grounding device and the neutral busbar of the lightning arrester.

b. The grounding grid in the fence leads to the adjacent fence grid or the main grounding grid of the site area which is designed by ECEPDI.

c. The grounding design of equipment (facilities), fences, terminal boxes, etc. is detailed in related installation volumes.

3. Construction Notes

a. The grounding grid inside 500kV filter bank fence adopts TJ-120 stranded copper conductor, with 0.8m laying depth and exothermic welding using for its connection. The grounding conductors crossing road and cable trench should be laid on the old-fashioned/backfill below the cushion.

b. The vertical grounding pole adopts Ø17.2mm, L=3000mm copper-coated steel rod, distance between any two vertical grounding poles should be larger than 6m, and all of them must be connected to grounding grid conductors using exothermic welding.

c. The transverse and longitudinal grounding bodies should be welded together, and the copper or copper-clad steel grounding body should be exothermic welding. Grounding grid should not coincide with the foundation by laying around from its side. Dimensions in each drawings are for reference, and can be fine-tuned according to the actual situation during construction.

d. All lightning arresters shall be provided with a centralized grounding pole connected to the grounding grid. All neutral busbars in the fence shall be reinforced with grounding. The vertical grounding electrode shall be added for the busbar and connected to the grounding grid.

e. All electrical equipment enclosures and bases, metal brackets, metal fences, metal doors, cable protection pipes, etc. shall be reliably welded to the grounding grid. The materials and schematic diagrams of grounding about those mentioned in the above are detailed in related installation volumes.

4. Others

This volume includes, but is not limited to, the above content. For other issues, please follow the international standards, national standards, industry standards, and construction acceptance standards.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

MLHVDC-LCS-CET-DD-EC-0405-001A

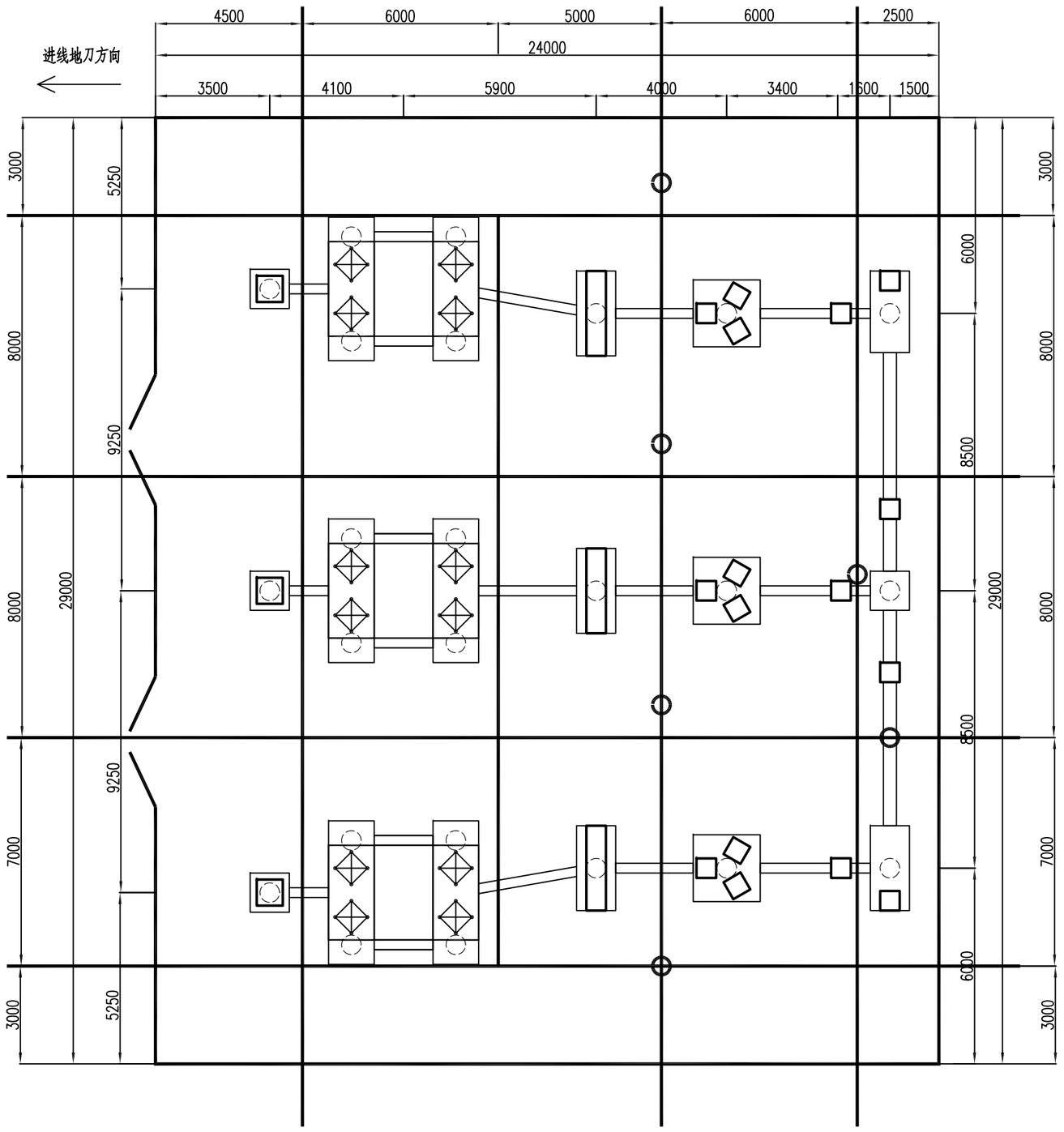
	姓名 Name	日期 Date
设计 Designed	范磊	2018.08.15
校核 Checked	赵葛城	2018.08.15
审核 Reviewed	田俊强	2018.08.15
批准 Approved	刘伟	2018.08.15

卷册施工说明

Description of Construction

比例 Scale	1:50	图号 DWG NO.	49-B60011S-D0405-001	版本 REV.	A

420X594



NOTES:

1. This drawing is about the grounding grid inside the SC fence. The horizontal grounding grid adopts TJ-120 stranded copper conductor and the laying depth is 0.8m. The vertical grounding pole uses ø17.2mm, L=3000mm copper-clad steel rods. All vertical grounding poles must be connected to the grounding grid, and the junctions should be exothermic welding.

2. The transverse and longitudinal grounding conductors should be welded together, and the copper or copper-clad steel grounding conductor should be exothermic welding. Grounding conductors should not coincide with the foundation by laying around from its side. Dimensions in each drawings are for reference, and can be fine-tuned according to the actual situation during construction.

3. All lightning arresters shall be provided with a centralized grounding pole connected to the grounding grid. All neutral busbars in the fence shall be reinforced with grounding. The vertical grounding electrode shall be added for the busbar and connected to the grounding grid.

4. All electrical equipment enclosures and bases, metal brackets, metal fences, metal doors, cable protection pipes, etc. shall be reliably welded to the grounding grid. The materials and schematic diagrams of grounding about those mentioned in the above are detailed in related installation volumes.

图例 Legend :

1、 垂直接地极 Vertical grounding pole

2、 水平接地网 Horizontal grounding grid

说明:

1、本图为SC围栏内辅助地网图，水平接地极采用TJ-120铜绞线，敷设深度为0.8m。垂直接地极采用ø17.2mm，L=3000mm铜覆钢棒，所有垂直接地极必须与辅助地网连接，连接处采用放热焊接。

2、横向和纵向接地体相互交叉处应焊接在一起，铜或者铜覆钢接地体采用放热焊接方式。地网不应与设备基础、构支架基础重合，遇到基础由侧面绕行敷设。图中标注的尺寸供参考，施工过程中可根据现场实际情况微调。围栏内地网引出至相邻围栏地网或至场区主地网。

3、围栏内所有避雷器均应设集中接地装置，集中接地装置就近与辅助接地网连接。围栏内所有中性点母线需加强接地，母线接地引下线增设垂直接地极，就近与辅助接地网连接。

4、围栏内所有电气设备外壳和底座、金属支架、金属围栏、金属门、电缆护管等均应与辅助地网可靠焊接。上述围栏内设备接地用材料及示意图详见各围栏设备安装卷册。

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 4 0 5 - 0 0 2 A

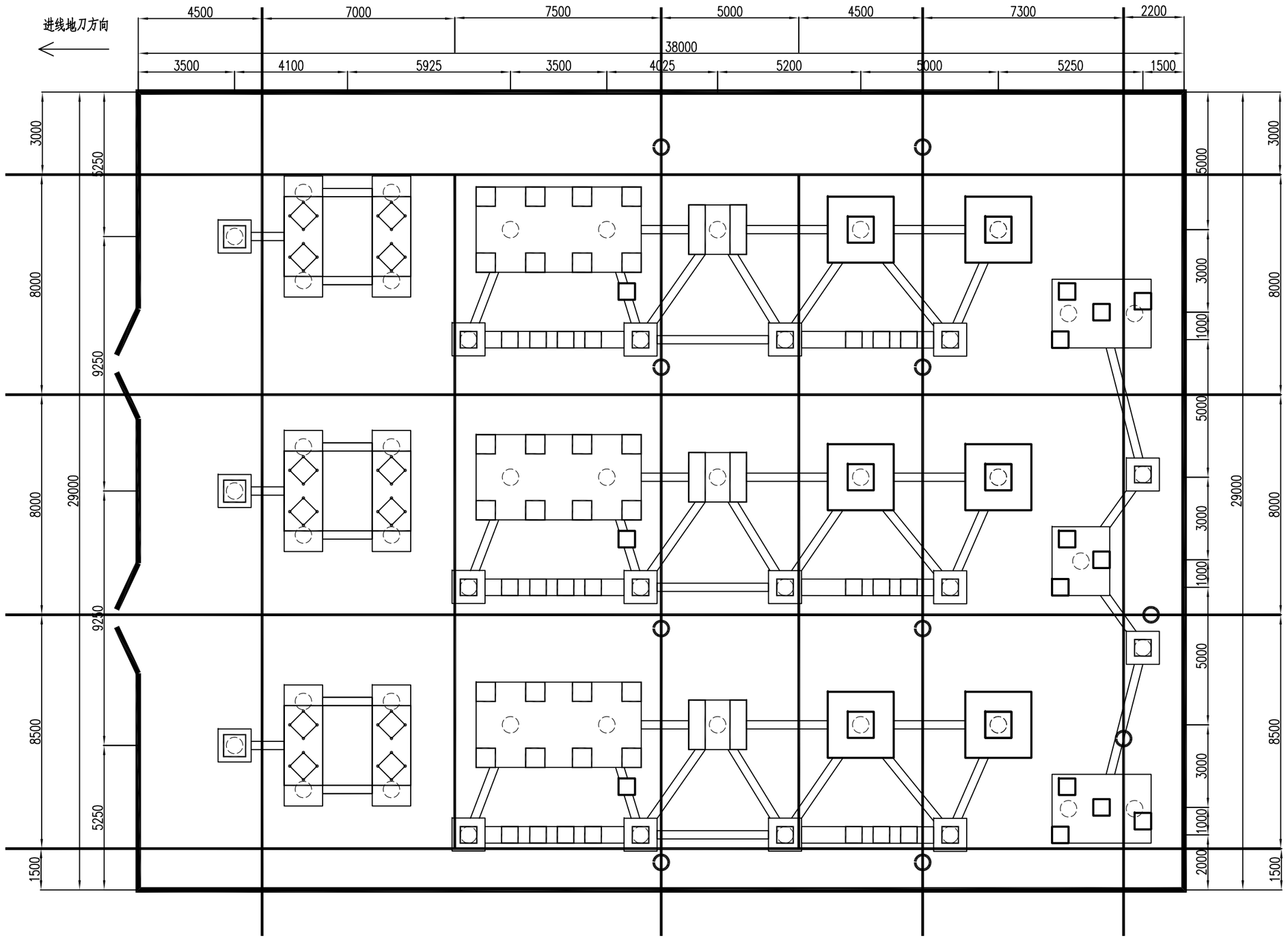
	姓名 Name	日期 Date
设计 Designed	范岩	2018.08.15
校核 Checked	李江 苏斌	2018.08.15
审核 Reviewed	田俊强	2018.08.15
批准 Approved	刘伟	2018.08.15

SC并联电容器小组地网布置图

SC Filter Bank Grounding Grid Plan

比例 Scale	1:150	图号 DWG NO.	49-B60011S-D0405-002	版本 REV.	A

420X594



NOTES:

1. This drawing is about the grounding grid inside the HP12/24 fence. The horizontal grounding grid adopts TJ-120 stranded copper conductor and the laying depth is 0.8m. The vertical grounding pole uses ø17.2mm, L=3000mm copper-clad steel rods. All vertical grounding poles must be connected to the grounding grid, and the junctions should be exothermic welding.

2. The transverse and longitudinal grounding conductors should be welded together, and the copper or copper-clad steel grounding conductor should be exothermic welding. Grounding conductors should not coincide with the foundation by laying around from its side. Dimensions in each drawings are for reference, and can be fine-tuned according to the actual situation during construction.

3. All lightning arresters shall be provided with a centralized grounding pole connected to the grounding grid. All neutral busbars in the fence shall be reinforced with grounding. The vertical grounding electrode shall be added for the busbar and connected to the grounding grid.

4. All electrical equipment enclosures and bases, metal brackets, metal fences, metal doors, cable protection pipes, etc. shall be reliably welded to the grounding grid. The materials and schematic diagrams of grounding about those mentioned in the above are detailed in related installation volumes.

图例 Legend :

1、 垂直接地极 Vertical grounding pole

2、 水平接地网 Horizontal grounding grid

说明:

1、本图为HP12/24围栏内辅助地网图，水平接地极采用TJ-120铜绞线，敷设深度为0.8m。垂直接地极采用ø17.2mm，L=3000mm铜覆钢棒，所有垂直接地极必须与辅助地网连接，连接处采用放热焊接。

2、横向和纵向接地体相互交叉处应焊接在一起，铜或者铜覆钢接地体采用放热焊接方式。地网不应与设备基础、构支架基础重合，遇到基础由侧面绕行敷设。图中标注的尺寸供参考，施工过程中可根据现场实际情况微调。围栏内地网引出至相邻围栏地网或至场区主地网。

3、围栏内所有避雷器均应设集中接地装置，集中接地装置就近与辅助接地网连接。围栏内所有中性点母线需加强接地，母线接地引下线增设垂直接地极，就近与辅助接地网连接。

4、围栏内所有电气设备外壳和底座、金属支架、金属围栏、金属门、电缆护管等均应与辅助地网可靠焊接。上述围栏内设备接地用材料及示意图详见各围栏设备安装卷册。

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 4 0 5 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	范岩	2018.08.15
校核 Checked	李延青	2018.08.15
审核 Reviewed	田俊强	2018.08.15
批准 Approved		2018.08.15
比例 Scale	1:150	图号 DWG NO.

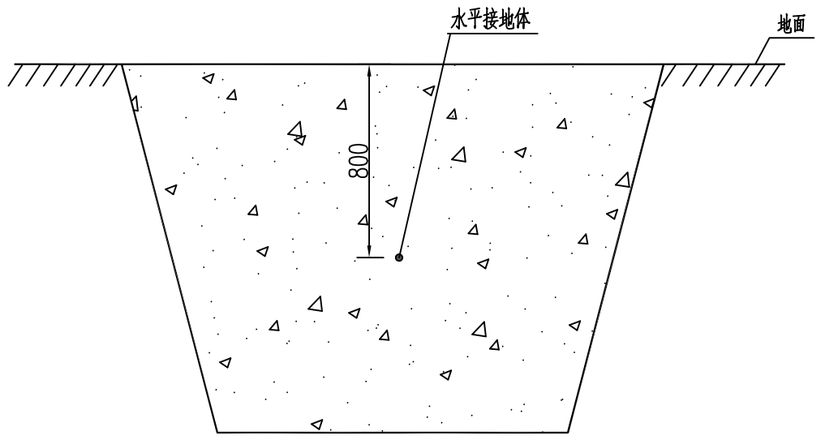
HP12/24交流滤波器小组地网布置图

HP12/24 AC Filter Bank Grounding Grid Plan

49-B60011S-D0405-003

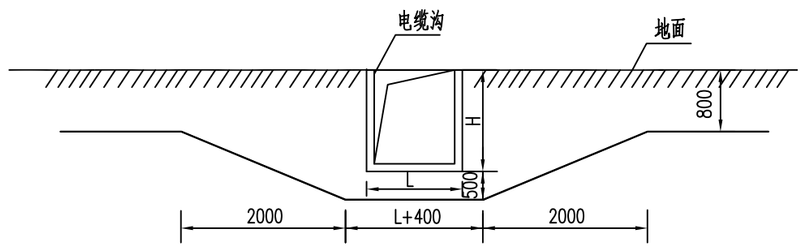
版本 REV. A

420X594



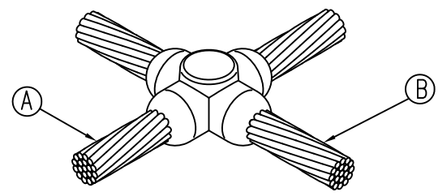
（图一）水平接地体敷设示意图

Figure 1. Horizontal Grounding Conductor Installation Diagram



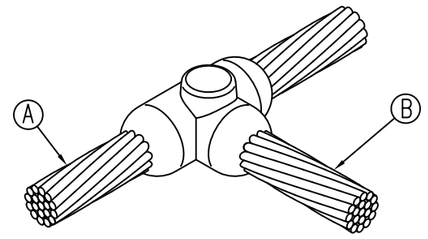
（图二）水平接地体过电缆沟敷设示意图

Figure 2 Horizontal grounding Conductor crossing cable trench Installation diagram



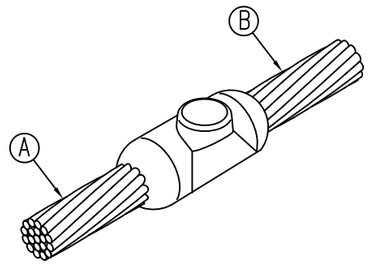
型式一
Type 1

A	B
120mm²	120mm²



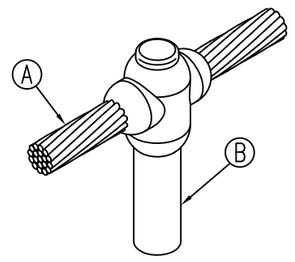
型式二
Type 2

A	B
120mm²	120mm²



型式三
Type 3

A	B
120mm²	120mm²



型式四
Type 4

A	B
120mm²	ø17.2mm

说明：

1、水平接地体的敷设在地下800mm。

2、水平接地体过电缆沟处，应由电缆沟下穿过，其敷设方式见图二。

NOTES:

1. The horizontal grounding conductor is laid 800mm underground.

2. Horizontal grounding conductor crossing the cable trench should be lied under it, which laying method is shown in Figure 2.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 4 0 5 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	范岩	2018.08.15
校核 Checked	李远 葛斌	2018.08.15
审核 Reviewed	田俊璟	2018.08.15
批准 Approved	张伟	2018.08.15
比例 Scale	1:100	图号 DWG NO.

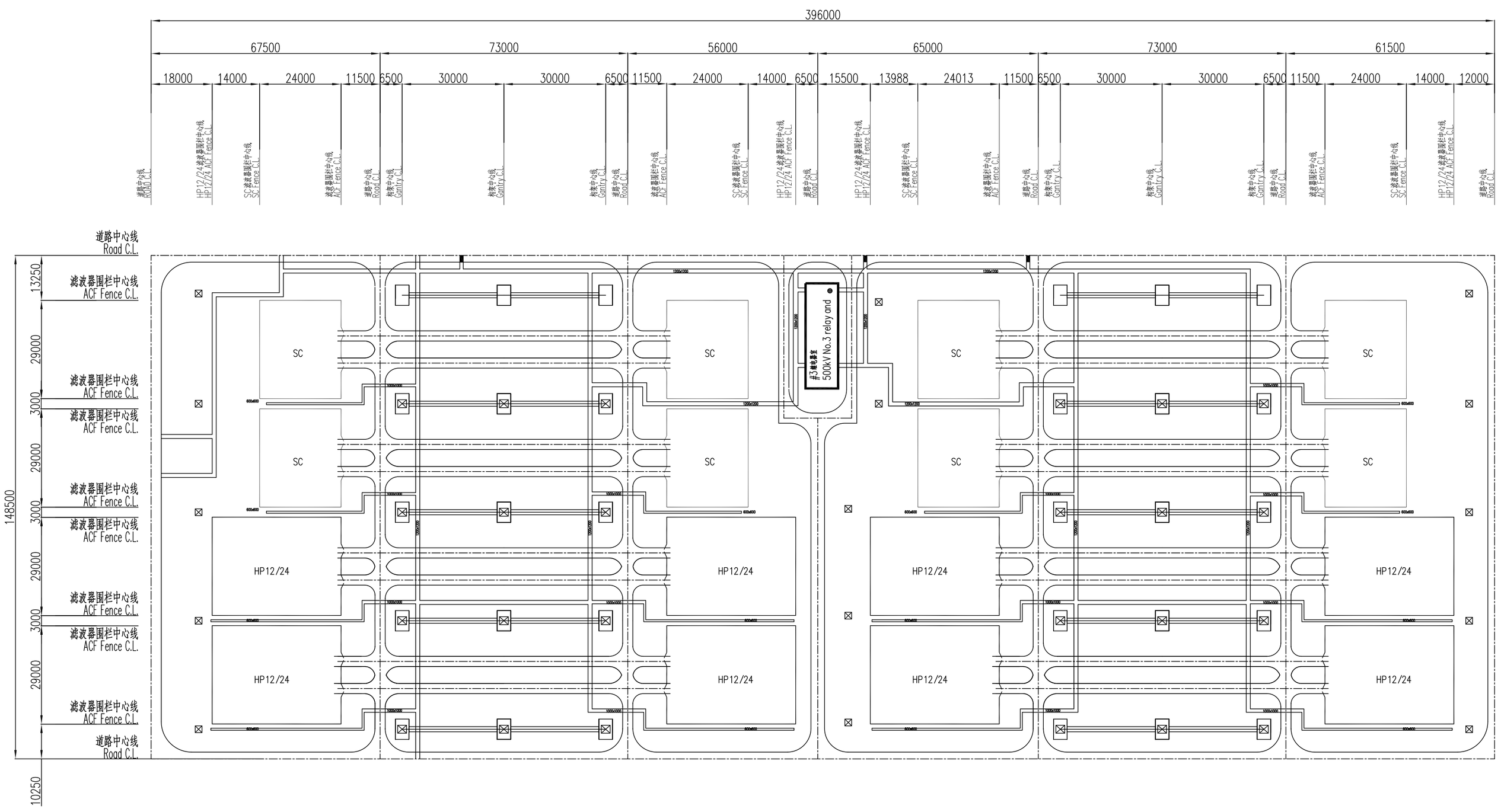
接地体敷设详图

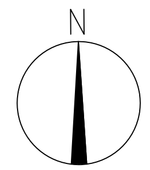
Detail Drawing of Grounding Conductor Laying

49-B60011S-D0405-004

版本 REV. A

420X594





Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER TECHNOLOGY EQUIPMENT CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 4 0 5 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	范之宏	2018.08.15
校核 Checked	李红 吉斌	2018.08.15
审核 Reviewed	田俊强	2018.08.15
批准 Approved	刘伟	2018.08.15
比例 Scale	1:750	图号 DWG NO. 49-B60011S-D0405-005

500kV交流滤波器区域围栏平面布置图

Layout of 500kV AC Filter Bank Fence

版本 REV. A

594X841

材料汇总表
Table of Material

序号 NO.	名 称 Name	规 范 Specification	单位 Unit	数量 Quantity	备 注 Remark
1	铜绞线 Stranded Copper Conductor	TJ-120	米 m	4840	水平接地网，甲供 Horizontal Grounding Grid, Contractor Supply
2	铜覆钢棒 Copper-Clad Steel Rod	ø17.2mm L=3m	根 Pcs	128	垂直接地极，甲供 Vertical Grounding Pole, Contractor Supply
3	放热焊点（形式一） Exothermic Welding(type 1)	TJ-120铜绞线,十字焊接 Type 1: A-120mm², B-120mm	个 Pcs	272	
4	放热焊点（形式二） Exothermic Welding(type 2)	TJ-120铜绞线,T字焊接 Type 2: A-120mm², B-120mm	个 Pcs	288	
5	放热焊点（形式三） Exothermic Welding(type 3)	TJ-120铜绞线,一字焊接 Type 3: A-120mm², B-120mm	个 Pcs	50	
6	放热焊点（形式四） Exothermic Welding(type 4)	TJ-120铜绞线与ø17.2mm镀铜钢棒 Type 4: A-150mm², B-ϕ17.2mm	个 Pcs	128	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR
国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 4 0 5 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	范磊	2018.08.15
校核 Checked	秦迎 葛斌	2018.08.15
审核 Reviewed	田俊强	2018.08.15
批准 Approved	张伟	2018.08.15

材料汇总表
Table of Material

比例 Scale	图号 DWG NO.	版本 REV.
1:50	49-B60011S-D0405-006	A

420X594